

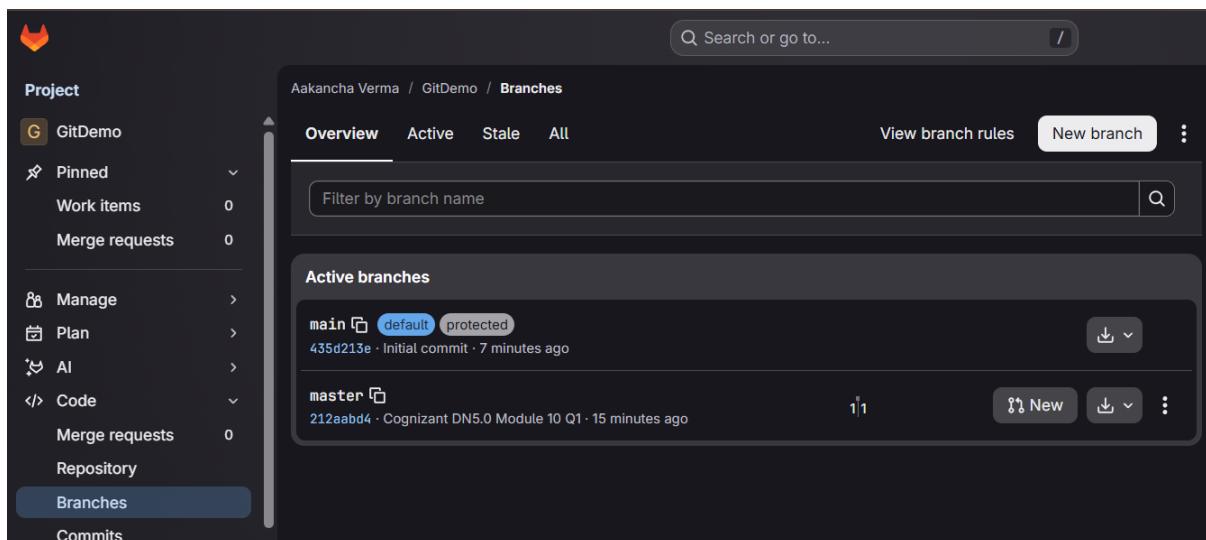
Module 10: GIT

1. Getting familiar with Git commands like git init, git status, git add, git commit, git push, and git pull.

Commands Used

```
git --version
git config --global user.name "Your Name"
git config --global user.email "your_email@gmail.com"
git config --list
notepad++
git config --global -e
git config --global core.editor "notepad++"
mkdir GitDemo
cd GitDemo
git init
ls -la
notepad++ welcome.txt
ls
cat welcome.txt
git status
git add welcome.txt
git commit
git remote add origin <repository-url>
git remote -v
git pull origin master
```

Output:



```

MINGW64:/d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo
aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ ls
welcome.txt

aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ cat welcome.txt
welcome to git.
This is my first Git Repository.
aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        welcome.txt

nothing added to commit but untracked files present (use "git add" to track)

aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git add welcome.txt

aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git commit
[master (root-commit) 212aabd] cognizant DN5.0 Module 10 Q1
 1 file changed, 2 insertions(+)
 create mode 100644 welcome.txt
cognizant cognizant cognizant
aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git remote add origin https://gitlab.com/aakanchavermavis/gitdemo.git

aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git remote -v
origin https://gitlab.com/aakanchavermavis/gitdemo.git (fetch)
origin https://gitlab.com/aakanchavermavis/gitdemo.git (push)

aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git pull origin master
info: please complete authentication in your browser...

aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git pull origin master
info: please complete authentication in your browser...
fatal: couldn't find remote ref master

aakan@AAKANCHA MINGW64 /d/documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git push origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 280 bytes | 140.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote:   https://gitlab.com/aakanchavermavis/gitdemo/-/merge_requests/new?merge_request%5Bsource_branch%5D=master
remote:
To https://gitlab.com/aakanchavermavis/gitdemo.git
 * [new branch]      master -> master

```

2. Understanding .gitignore

Complete Command List

```

cd /d/GitDemo
pwd
git status
touch error.log
mkdir logs
cd logs
touch app.log
cd ..
ls
ls logs
notepad++ .gitignore
cat .gitignore
git status
git add .gitignore
git commit -m "Added gitignore file"
git status

```

```
git push origin master
```

Outputs:

```
MINGW64:/d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo
$ notepad ++ error.log

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ notepad++ error.log
AAA
aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ mkdir logs

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ cd logs

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo/logs (master)
$ notepad++ app.log

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo/logs (master)
$ cd ..

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ ls
error.log  logs/  welcome.txt

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ ls logs
app.log

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ notepad++ .gitignore

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ cat .gitignore
*.log
logs/

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git add .gitignore

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git commit -m "Added gitignore file"
[master a0ca014] Added gitignore file
1 file changed, 2 insertions(+)
create mode 100644 .gitignore

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

aakan@AAKANCHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 300 bytes | 300.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote:   https://gitlab.com/aakanchavermavis/gitdemo/-/merge_requests/new?merge_request%5Bsource_branch%5D=master
remote:
To https://gitlab.com/aakanchavermavis/gitdemo.git
   212aabd..a0ca014  master -> master
```

Aakancha Verma / GitDemo / Repository

 You pushed to master 17 seconds ago

Create merge request

Files

master gitdemo

Search files (*.vue, *.rb...)

gitdemo

+ Find file Code

 .gitignore

 welcome.txt

 Added gitignore file
Aakancha Verma authored 54 se... a0ca014 History

Name	Last commit	Last update
 .gitignore	Added gitignore file	54 seconds ago
 welcome.txt	Cognizant DN5.0 ...	39 minutes ago

3. Understanding Branching and Merging

Commands Used

```
cd /d/GitDemo
pwd
git branch
git branch GitNewBranch
git branch -a
git checkout GitNewBranch
notepad++ branch.txt
git status
git add branch.txt
git commit -m "Added branch.txt in GitNewBranch"
git status
git checkout master
git diff master GitNewBranch
git difftool master GitNewBranch
git merge GitNewBranch
git log --oneline --graph --decorate
git branch -d GitNewBranch
git branch
git status
git push origin master
```

Outputs:

The screenshot displays the GitHub interface for a repository named 'gitdemo'. At the top, a notification bar indicates a successful push to the 'master' branch 1 minute ago, with a 'Create merge request' button. Below this, the 'Files' tab is active, showing a search bar and a list of files: '.gitignore', 'branch.txt', and 'welcome.txt'. To the right, a commit summary for 'Added branch.txt in GitNewBranch' is shown, including the commit hash 'dc82ef5a' and a 'History' button. At the bottom, a table lists the files and their commit details.

Name	Last commit	Last update
.gitignore	Added gitignore file	18 minutes ago
branch.txt	Added branch.txt...	5 minutes ago
welcome.txt	Cognizant DN5.0 ...	57 minutes ago

```

MINGW64:/d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..0bd651f
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1,3 @@
+This file is created in GitNewBranch.
+
+This is my branching assignment.
\ No newline at end of file

aakan@AAKANCHHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git difftool master GitNewBranch

This message is displayed because 'diff.tool' is not configured.
See 'git difftool --tool-help' or 'git help config' for more details.
'git difftool' will now attempt to use one of the following tools:
opendiff kdiff3 tkdiff meld kompare gvimdiff diffuse diffmerge ecmerge p4merge araxis bc codecompare smmerge emerge vimdiff nvimdiff

Viewing (1/1): 'branch.txt'
Launch 'vimdiff' [Y/n]? n

aakan@AAKANCHHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git branch
  GitNewBranch
* master

aakan@AAKANCHHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git merge GitNewBranch
Updating a0ca014..dc82ef5
Fast-forward
  branch.txt | 3 +++
  1 file changed, 3 insertions(+)
  create mode 100644 branch.txt

aakan@AAKANCHHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git log --oneline --graph --decorate
* dc82ef5 (HEAD -> master, GitNewBranch) Added branch.txt in GitNewBranch
* a0ca014 (origin/master) Added gitignore file
* 212aabd Cognizant DNS.0 Module 10 Q1

aakan@AAKANCHHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was dc82ef5).

aakan@AAKANCHHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git branch
* master

aakan@AAKANCHHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

aakan@AAKANCHHA MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 396 bytes | 198.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote:   https://gitlab.com/aakanchavermavis/gitdemo/-/merge_requests/new?merge_request%5Bsource_branch%5D=master
remote:
To https://gitlab.com/aakanchavermavis/gitdemo.git
  a0ca014..dc82ef5 master -> master

```

4. Resolving Conflict during Merge

Commands Used

```

cd /d/GitDemo
$ git checkout master
git status
git branch GitWork
git checkout GitWork
notepad++ hello.xml
git status
git add hello.xml
git commit -m "Added hello.xml in GitWork"
git checkout master
notepad++ hello.xml
git status
git add hello.xml

```

```

git commit -m "Added hello.xml in master"
git log --oneline --graph --decorate --all
git diff master GitWork
git difftool master GitWork
git merge GitWork
notepad++ hello.xml
git add hello.xml
git commit -m "Resolved merge conflict"
git status
notepad++ .gitignore
git add .gitignore
git commit -m "Updated gitignore for backup files"
git branch
git branch -d GitWork
git log --oneline --graph --decorate
git push origin master

```

Outputs:

Aakancha Verma / GitDemo / Repository

You pushed to **master** 1 minute ago

Create merge request

Files
master gitdemo
Search files (*.vue, *.rb...)
.gitignore
branch.txt
hello.xml
welcome.txt

Updated gitignore for back...
Aakancha Verma authored 2 min...
5991050a
History

Name	Last commit	Last update
.gitignore	Updated gitignor...	2 minutes ago
branch.txt	Added branch.txt...	18 minutes ago
hello.xml	Resolved merge ...	3 minutes ago
welcome.txt	Cognizant DN5.0 ...	1 hour ago

```
MINGW64:/d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo
GitWork
* master

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master|MERGING)
$ notepad++ hello.xml

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master|MERGING)
$ git add hello.xml

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master|MERGING)
$ git commit -m "Resolved merge conflict"
[master d3172db] Resolved merge conflict

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ notepad++ .gitignore

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git add .gitignore

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git commit -m "Updated gitignore for backup files"
[master 5991050] Updated gitignore for backup files
1 file changed, 3 insertions(+), 1 deletion(-)

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git branch
GitWork
* master

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git branch -d GitWork
Deleted branch GitWork (was 84cdc07).

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git log --oneline --graph --decorate
* 5991050 (HEAD -> master) Updated gitignore for backup files
* d3172db Resolved merge conflict
| \
| * 84cdc07 Added hello.xml in GitWork
* | 53e3112 Added hello.xml in master
| /
* dc82ef5 (origin/master) Added branch.txt in GitNewBranch
* a0ca014 Added gitignore file
* 212aabd Cognizant DNS.0 Module 10 Q1

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git push origin master
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 12 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 1.22 KiB | 249.00 KiB/s, done.
Total 12 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote: https://gitlab.com/aakanchavermavis/gitdemo/-/merge_requests/new?merge_request%5Bsource_branch%5D=master
remote:
To https://gitlab.com/aakanchavermavis/gitdemo.git
dc82ef5..5991050 master -> master
```

5. Understanding clean up and push back to remote Git

Commands Used

```
cd /d/GitDemo
git checkout master
git status
git branch
git branch -a
git pull origin master
git push origin master
git log --oneline --graph --decorate
git status
```

Outputs:

```
MINGW64:/d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo
$ git branch -d GitWork
Deleted branch GitWork (was 84cdc07).

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git log --oneline --graph --decorate
* 5991050 (HEAD -> master) Updated gitignore for backup files
* d3172db Resolved merge conflict
| \
| * 84cdc07 Added hello.xml in GitWork
* | 53e3112 Added hello.xml in master
| /
* dc82ef5 (origin/master) Added branch.txt in GitNewBranch
* a0ca014 Added gitignore file
* 212aabd Cognizant DNS.0 Module 10 Q1

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git push origin master
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 12 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 1.22 KiB | 249.00 KiB/s, done.
Total 12 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote:   https://gitlab.com/aakanchavermavis/gitdemo/-/merge_requests/new?merge_request%5Bsource_branch%5D=master
remote:
To https://gitlab.com/aakanchavermavis/gitdemo.git
   dc82ef5..5991050  master -> master

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git branch
* master

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git branch -a
* master
  remotes/origin/master

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git pull origin master
From https://gitlab.com/aakanchavermavis/gitdemo
   * branch      master       -> FETCH_HEAD
Already up to date.

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git push origin master
Everything up-to-date

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git log --oneline --graph --decorate
* 5991050 (HEAD -> master, origin/master) Updated gitignore for backup files
* d3172db Resolved merge conflict
| \
| * 84cdc07 Added hello.xml in GitWork
* | 53e3112 Added hello.xml in master
| /
* dc82ef5 Added branch.txt in GitNewBranch
* a0ca014 Added gitignore file
* 212aabd Cognizant DNS.0 Module 10 Q1

aakan@AAKANCH A MINGW64 /d/Documents/Internships and Hackathons/Cognizant Java Digital Nurture/Coding/Module 10/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean
```


✓ You pushed to `master` 8 minutes ago



Create merge request

Files

master gitdemo

Search files (*.vue, *.rb...)



gitdemo



Find file

Code



.gitignore
branch.txt
hello.xml
welcome.txt



Updated gitignore for back...
Aakancha Verma authored 9 min...

5991050a



History

Name	Last commit	Last update
.gitignore	Updated gitignor...	9 minutes ago
branch.txt	Added branch.txt...	25 minutes ago
hello.xml	Resolved merge ...	10 minutes ago
welcome.txt	Cognizant DN5.0 ...	1 hour ago